From Practical Life to Independence:

A Research Proposal on Montessori Practical Life Activities and Independence in Children Aged 4-5

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Abstract

This research proposal demonstrates the research process through a focused study of Montessori Practical Life activities and independence in children aged 4-5. The literature indicates that Montessori education is associated with positive academic and nonacademic outcomes, including executive function, social understanding, and mastery orientation; however, most rigorous studies evaluate the Montessori program as a whole rather than isolating Practical Life activities. Practical Life is theoretically important because it gives young children repeated opportunities to care for themselves, care for the environment, complete purposeful sequences, correct mistakes, and return materials independently. To address the narrower evidence gap, the proposed study uses a quasi-experimental pretest-posttest comparison design with approximately 40 children in two preschool classrooms over eight weeks. The independent variable is structured participation in Practical Life activities, and the primary dependent variable is observable independence across self-care, environmental care, task completion, and help-seeking behaviors. Quantitative observations would be supplemented by brief field notes. Planned analyses include descriptive statistics, baseline equivalence checks, change scores, an analysis of posttest independence controlling for baseline, and examination of associations between participation frequency and improvement. The proposal emphasizes ethical observation, consistent operational definitions, transparent data organization, and cautious interpretation. It illustrates how a literature review leads logically to a research gap, research questions, hypotheses, methods, and an analysis plan.

Keywords: Montessori, Practical Life, independence, preschool, early childhood education, research process

From Practical Life to Independence: A Research Proposal

Introduction and the Research Process

A strong research project begins with a focused problem rather than with data collection. The researcher first identifies a topic, reviews what is already known, defines the concepts that will be studied, and looks for a meaningful gap or unresolved question. The literature review then becomes the bridge between the broad topic and the specific research question. After a question is stated, the researcher selects a design, identifies participants and measures, plans how data will be organized and analyzed, and decides in advance what kinds of conclusions the design can and cannot support. This sequence creates research coherence because each stage follows logically from the stage before it.

The present proposal applies that process to early childhood Montessori education. The topic is the relationship between Montessori Practical Life activities and independence among children aged 4-5. This is a useful research problem because independence is a central developmental goal in Montessori early childhood practice, yet it can be defined too broadly unless it is translated into observable behaviors. In this proposal, independence is not treated as a personality label. It is operationalized as the child's ability to initiate age-appropriate tasks, complete a short sequence with limited adult assistance, manage materials, respond to minor errors, care for the environment, and return materials after use.

Literature Review

Practical Life and the Montessori Concept of Independence

In Montessori early childhood environments, Practical Life activities use familiar, purposeful tasks such as pouring, transferring, washing, dressing, food preparation, cleaning, and

caring for plants. Their educational value lies not only in successful task completion but also in the sequence of actions required: selecting materials, carrying them carefully, following an orderly process, noticing errors, restoring the environment, and returning the materials. Dorer (2018) describes independence as a major developmental aim of Montessori education and argues that a strong Practical Life program should be paired with restrained adult intervention. This principle is important for the present study because independence cannot develop if adults routinely complete manageable steps for the child.

Practical Life also connects physical action with attention and self-regulation. Activities are usually designed at child scale, arranged in a predictable order, and presented slowly. A child who pours water, wipes a spill, folds a cloth, or returns a tray receives immediate information from the environment about whether the action was successful. This creates opportunities for self-correction without requiring constant verbal evaluation from an adult. Conceptually, these experiences may support autonomy, sustained attention, fine motor coordination, responsibility, and behavioral regulation. However, these mechanisms should be treated as plausible pathways rather than assumed outcomes unless they are measured directly.

Evidence for Montessori Outcomes

Empirical research on Montessori education provides a broader context for this proposal. Lillard and Else-Quest (2006) compared children in a public Montessori program with children who had lost a school admission lottery and attended other programs. Their results provided early evidence that Montessori education may be associated with academic and social advantages. Later, Lillard (2012) examined preschool children in classic Montessori, supplemented Montessori, and conventional programs. Children in high-fidelity classic Montessori classrooms showed stronger school-year gains on several measures, including

executive function and social problem solving. That study is especially relevant because it highlights implementation fidelity: the label "Montessori" alone does not guarantee that classrooms provide the same learning experiences.

A longitudinal lottery-based study by Lillard et al. (2017) followed children across three preschool years. The Montessori group showed stronger growth on several outcomes, including academic achievement, social understanding, and mastery orientation, and four-year-old Montessori children showed an advantage on executive function. More recently, a systematic review by Randolph et al. (2023) synthesized 32 studies that met baseline-equivalence requirements. Across those studies, Montessori education had positive average effects on both academic and nonacademic outcomes, with nonacademic outcomes including executive function, social skills, creativity, and children's experience of school. The review also found that effects varied across study design and setting, reinforcing the need to specify exactly what form of Montessori implementation is being investigated.

The evidence base continues to develop. In a national randomized controlled trial across 24 public Montessori schools, Lillard et al. (2025) followed 588 children who entered competitive lotteries at age three. At the end of kindergarten, children offered Montessori seats showed advantages on reading, short-term memory, executive function, and social understanding, although the study evaluated public Montessori preschool as a complete model rather than Practical Life as an isolated component. This distinction is important: evidence that a whole educational system is effective does not establish which particular component produces each outcome.

Limitations in the Existing Literature and the Research Gap

Marshall (2017) notes that the Montessori evidence base has historically been limited by small samples, inconsistent implementation, selection effects, and variation in outcome measures. Although later studies and systematic reviews have strengthened the evidence, a specific gap remains for the present topic. Most rigorous studies compare entire Montessori programs with other forms of schooling. They therefore combine Practical Life with Montessori language, mathematics, sensorial materials, multi-age grouping, long work cycles, child choice, teacher behavior, and other features. It is difficult to infer the independent contribution of Practical Life from such designs.

The proposed study addresses this narrower gap by measuring participation in Practical Life activities and observable independence within a defined age group. It does not attempt to prove that Practical Life alone causes all broader Montessori outcomes. Instead, it asks a more focused and testable question: whether increased, structured opportunities to perform age-appropriate Practical Life tasks are associated with measurable gains in independent classroom behavior over a short period. This focus improves construct clarity and keeps the proposed claims proportional to the design.

Statement of the Research Problem

Montessori educators frequently describe Practical Life as a foundation for independence, but the relationship is often discussed philosophically or as part of the entire Montessori model rather than measured directly. Teachers therefore have limited component-specific evidence about how much observable independence changes when children aged 4-5 receive systematic opportunities to choose, complete, correct, and restore Practical Life work with restrained adult

assistance. A focused classroom study can help connect Montessori concepts to observable behaviors while also demonstrating a clear research process.

Research Questions and Hypotheses

The study is guided by three research questions:

- RQ1. Do children aged 4-5 who participate in a structured eight-week Montessori Practical Life program show greater improvement in observable independence than children who continue with their usual preschool routine?
- RQ2. Which dimensions of independence - self-care, care of the environment, task completion, and appropriate help-seeking - show the greatest change over the study period?
- RQ3. Within the Practical Life group, is frequency of independent participation associated with greater improvement in independence scores?

The primary directional hypothesis (H1) is that, after accounting for baseline independence, children in the structured Practical Life condition will have higher posttest independence scores than children in the comparison condition. A secondary hypothesis (H2) is that greater frequency of independently completed Practical Life activities will be positively associated with improvement in independence scores. The null hypotheses state that there will be no statistically reliable group difference in adjusted posttest independence and no reliable association between participation frequency and improvement.

Research Method

Design

A quasi-experimental pretest-posttest comparison design is proposed. Two existing preschool classrooms serving children aged 4-5 would participate. One classroom would implement the structured Practical Life condition for eight weeks, while the other would continue its usual classroom routine during the same period. Using existing classrooms makes the study feasible, but it also means the children are not individually randomized. For that reason, the design should be described as quasi-experimental rather than as a randomized controlled trial, and causal conclusions should remain cautious.

Participants and Setting

The target sample is approximately 40 children aged 4-5, with about 20 children in each classroom. Inclusion criteria are enrollment in the participating classroom, parent or guardian consent, and sufficient attendance to provide both baseline and posttest observations. The setting should be a preschool with comparable schedules, teacher experience, classroom size, and daily routines across the two groups. Basic demographic information, age in months, attendance, and prior Montessori exposure should be recorded because these variables may influence baseline independence or participation.

Variables and Operational Definitions

The independent variable is exposure to the structured Practical Life condition. During a protected work period, children in this condition would have daily access to sequenced activities such as pouring water, wiping a table, arranging materials, simple self-care, and caring for plants. Teachers would demonstrate slowly, invite participation, and avoid completing

manageable steps for the child. The comparison group would continue its normal preschool program without the added structured Practical Life sequence.

The primary dependent variable is observable independence. A researcher-developed observation rubric would score four domains: (a) self-care, (b) care of the environment and materials, (c) task initiation and completion, and (d) appropriate help-seeking and response to minor errors. Each domain could be scored from 0 to 3, where 0 indicates full adult completion, 1 indicates substantial assistance, 2 indicates one or two prompts, and 3 indicates independent completion. A total score would range from 0 to 12. Operational definitions and examples must be written before data collection so that different observers interpret behaviors consistently.

Instruments and Observation Procedures

The main instrument would be a structured independence observation rubric administered at baseline and again during the final week. Each child would be observed during comparable classroom periods on at least two occasions at each measurement point. A second trained observer should independently score a subset of observations so that inter-rater agreement can be examined. In addition to rubric scores, brief field notes would record contextual information such as the specific activity, whether the child encountered a minor difficulty, and the type of adult prompt, if any. These notes are supplementary and should not replace the predefined quantitative scoring rules.

For the intervention group, teachers would also maintain a simple participation log. The log would record the date, activity category, whether the child chose or accepted the activity, whether the activity was completed, and the highest level of assistance required. The log is not a grade. Its purpose is to measure exposure so that RQ3 can examine whether children who engage more frequently also show larger gains.

Procedure

Week 1 would be used for consent, observer training, and baseline measurement. During Weeks 2-7, the Practical Life group would receive consistent daily access to the selected activities, with teachers following the same presentation and non-intervention guidelines. The comparison classroom would continue its normal routine. Week 8 would be used for posttest observations. To reduce measurement bias, observers scoring posttest behavior should be trained to use behavioral criteria rather than general impressions, and, where feasible, the primary scorer should not be told the child's baseline total while rating the posttest observation.

Ethical Considerations

Because the participants are young children, parent or guardian informed consent and age-appropriate child assent are essential. Participation should not reduce access to normal educational experiences, and children should be free to decline an activity without penalty. Data should be identified by study codes rather than children's names, and any video used for reliability scoring would require explicit additional permission and secure storage. The study should be reviewed under the institution's applicable research ethics procedures before data collection begins.

Data Organization and Interpretation

Data Organization

Data should be organized before analysis in a rectangular dataset in which each row represents one child and each column represents one defined variable. A separate codebook should list every variable name, description, allowed value, and missing-data code. This prevents later ambiguity and makes the analysis reproducible. Raw observation sheets should be retained

separately from the cleaned analysis file so that data entry can be checked without overwriting the original records.

Variable	Example coding	Purpose
ID	P001-P040	Anonymous participant identifier
Group	0 = comparison; 1 = Practical Life	Independent variable
AgeMonths	48-71	Participant characteristic/covariate
PreIndependence	0-12	Baseline outcome
PostIndependence	0-12	Posttest outcome
ChangeScore	Post - Pre	Descriptive improvement
ParticipationCount	0-n activities	Exposure measure for RQ3
AttendancePct	0-100	Context/covariate

Quantitative Analysis Plan

Analysis would begin with data screening: checking missing values, impossible scores, duplicate IDs, and outliers. Descriptive statistics would summarize age, attendance, baseline independence, posttest independence, and participation frequency. Baseline group differences should be examined before testing the primary hypothesis. Internal consistency of the multi-item rubric may be summarized if the domains are intended to form a total score, and agreement between two observers should be reported for the subset of double-scored observations.

For RQ1, the clearest primary analysis would compare posttest independence between groups while statistically controlling for baseline independence, provided model assumptions are reasonably met. A simpler classroom assignment could also report mean change scores for each group with confidence intervals and an effect size. The emphasis should be on the magnitude and uncertainty of the difference, not only on whether a p value crosses .05. For RQ2, change can be summarized separately for the four independence domains. For RQ3, the relationship between participation count and independence change can be examined with a correlation or a simple regression within the Practical Life group. If participation counts are highly skewed or the sample is small, a rank-based correlation may be more appropriate.

Interpretation Principles

Interpretation must remain tied to the design. A statistically significant adjusted group difference would support the hypothesis that the structured Practical Life condition is associated with greater short-term gains in observable independence in these classrooms. It would not justify the stronger claim that Practical Life universally causes independence or that the effect will persist into later childhood. Likewise, a positive participation-gain association would be consistent with a dose-response pattern, but children who choose more activities may already differ in motivation, confidence, or teacher relationships. Those alternative explanations should be discussed rather than ignored.

Qualitative field notes can add context to the quantitative scores by identifying recurring patterns, such as children independently correcting a spill, returning a tray without reminders, or requesting help only after attempting a task. These notes should be organized by predefined categories and used to illustrate patterns, not to manufacture themes that are unsupported by the observations. Because this document is a proposal, no numerical results are reported or invented. Actual interpretation would occur only after the planned data have been collected and checked.

Research Coherence and Application of Concepts

The coherence of the proposal comes from alignment. The literature identifies independence as an important Montessori goal and suggests that Montessori environments can support nonacademic development, while also showing that whole-program studies cannot isolate the effect of Practical Life. That gap leads directly to the research questions. The questions then determine the variables: structured Practical Life exposure, observable independence, independence subdomains, and participation frequency. The variables determine

the observation rubric, participation log, design, and analysis. Finally, the limits of the quasi-experimental design determine the cautious language used for interpretation.

Several research concepts are applied explicitly. Construct validity is addressed by defining independence behaviorally rather than as a vague trait. Reliability is addressed through standardized scoring and double-coded observations. Internal validity is strengthened by baseline measurement, comparable observation periods, and statistical adjustment, while remaining limited by nonrandom classroom assignment. External validity is limited because a small number of classrooms may not represent all Montessori or preschool settings. Implementation fidelity matters because previous research shows that Montessori outcomes can vary with how closely the program is implemented (Lillard, 2012). These concepts ensure that the method is not merely a list of procedures but a reasoned plan for producing interpretable evidence.

Limitations and Delimitations

The proposed study has several limitations. First, two intact classrooms cannot eliminate selection effects or teacher effects. Second, a researcher-developed independence rubric would require careful pilot testing and reliability evidence. Third, eight weeks may capture short-term behavioral change but not long-term development. Fourth, observer presence may temporarily alter children's behavior. Fifth, the intervention contains several Practical Life activities, so the study cannot determine whether pouring, cleaning, self-care, or plant care is individually responsible for change. The study is deliberately delimited to children aged 4-5 and to independence during preschool routines; it does not attempt to measure academic achievement, creativity, or long-term personality development.

Conclusion

This proposal demonstrates how to construct a coherent research process from literature review to research question, hypothesis, method, data organization, and interpretation. Existing evidence suggests that Montessori education can support important academic and nonacademic outcomes, and Montessori theory places Practical Life at the center of children's movement toward independence. At the same time, most rigorous studies evaluate Montessori as a complete educational model. A focused, carefully defined classroom study of Practical Life and observable independence therefore addresses a reasonable gap without overstating what prior evidence has established. The strongest version of the proposed study would use clear operational definitions, consistent observation procedures, documented implementation, transparent data management, and cautious conclusions that match the quasi-experimental design.

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Appendix A

Rubric Alignment Checklist

Rubric criterion	**Where it is addressed**
Title and Abstract	Specific research title; structured abstract summarizes problem, literature, method, and analysis.
Research Questions	Three focused research questions plus directional and null hypotheses.
Research Method	Design, participants, variables, instruments, procedure, and ethics are specified.
Data Organization and Interpretation	Codebook structure, example variables, analysis plan, and interpretation limits are explained.
Depth and Focus	Each section develops one research function and remains centered on Practical Life and independence.
Research Coherence	Literature gap leads to questions; questions lead to variables, measures, analysis, and bounded conclusions.
Concepts and Application	Construct validity, reliability, internal validity, external validity, fidelity, and operationalization are applied.
Sentence Structure	Formal academic prose uses complete sentences, transitions, and varied structure.
Mechanics and Grammar	Edited for standard academic English, punctuation, capitalization, and agreement.
Format, Citations and References	APA-style in-text citations and an alphabetized reference list with DOI links where available.